Shadows of Almia Showdown

Complete project brief: what the game is, every gameplay system with its numbers, the code, the island and its generator, online play and operations, the work done, the conventions, and what is open. Written 27 October 2026 for handing the whole context to another assistant.

1. What the project is

Shadows of Almia Showdown is a browser Pokémon battle royale built by Wesley Millet with Claude (Anthropic) as the engineer. Eight trainers land on a 320 x 224 tile island, catch and train wild Pokémon, fight each other in one-on-one trainer battles in real time (moves on cooldowns, aimed with the mouse), and the last trainer standing wins while a creeping darkness shrinks the safe ground toward the Shadow Crystal at the centre, where Darkrai finally appears. It is a non-commercial fan game: Pokémon sprites from the PMD SpriteCollab project, human sprites from Pokémon Ranger: Shadows of Almia, move effects from Pokémon Ranger: Guardian Signs, and the island painted with The Fan-tasy Tileset (a purchased pixel-art pack by Valerio Colonna). Keep it free, no ads, no payments.

Where	Address / location
Live online server (Fly.io)	https://almia-showdown.fly.dev
Offline demo (GitHub Pages, branch main)	https://wesmillet.github.io/Shadows-of-Almia-Showdown/
Repository	github.com/wesmillet/Shadows-of-Almia-Showdown
Branches	**online**: server plus online client, deployed to Fly. **main**: the offline demo (no server/ folder, no online UI), deployed to GitHub Pages. Every change is committed on online and cherry-picked to main (server-only files dropped), then main is merged back into online with the ours strategy.
Deploying	GitHub Actions workflow "Launch online server" (launch-online.yml, ref main, input size=friends) deploys the live server; "smoke-online.yml" runs an end-to-end smoke test and writes its result to a GitHub issue; pushes to main rebuild the Pages demo (deploy.yml); ci.yml runs types, tests, both builds and the server checks on every push.

2. Technology and architecture

- **Client**: TypeScript, Vite, one HTML entry. Canvas 2D for the world, a hand-written WebGL pass for lighting (warm daylight tint that shifts cold over the match, the darkness ring, additive lights, vignette, depth-of-field blur at the top of the screen; a Canvas 2D approximation without WebGL, also the "simple graphics" option). Sprites draw at an integer 3x with nearest filtering; the ground scales with the window. Pokémon are drawn at their Pokédex size against 1.6 m trainers (small ones lifted on a ramp so nothing is under 0.9 m, long ones measured along their length, giants capped at 6 m).
- **Simulation** (src/sim): deterministic 60 Hz fixed steps with render interpolation; the match (match.ts), battle rules (battle.ts), AI (ai.ts), catching, items, evolution, darkness, fishing, mega evolution, navigation, movement, party formation, seats, towns, townsfolk, visitors. The battle system emits events (cast, resolve, hit, status, faint, caught, scene-fx) that the match screen turns into effects, hit-stop, shake and floating numbers. The same simulation runs offline in the browser and authoritatively on the server.
- **Server** (server/, branch online): Node 22, TypeScript via tsx and esbuild, ws WebSockets, SQLite through node:sqlite. One program serves the pages and art, runs every match in its own worker thread (remoteRoom.ts, roomWorker.ts, one loaded and waiting), the matchmaking lobby (matchmaker.ts), global chat, accounts and saves (accounts.ts), trading, moderation, ratings, replays, backups, telemetry, events, profile and privacy pages, an admin CLI. Browsers send keys, cursor and commands; 20 times a second each player receives the changes to the part of the island they can see (view.ts, src/net/sync.ts; undefined fields travel as deletions) and mirrors the match (src/net/mirror.ts) with client-side prediction for movement and lag-compensated aiming.
- **Data** (data/*.json): species.json (322 species with types, base stats, learnsets, evolutions, sizes, starter tiers), moves.json (175 moves, each with an animation recipe), items/items.json (64 items), items/tms.json (38 TMs),

items/loot.json (item ball tiers), types.json, spawns.json (wild tables, rotation pools, bosses, starter tiers, trainer pool, mega lines), shops.json (stalls, prices, money, the Castle Champion), fishing.json, shiny.json (palette swaps), characters.json (NPC roles), config.json (every tunable).

- **Assets** (assets/): 324 Pokémon sprite folders (SpriteCollab AnimData.xml plus Anim, Shadow and Offsets sheets; shiny and shadow forms as separate folders), portraits, 23 ranger sheets with a manifest (ten Ranger Union members, thirteen townsfolk), 256 Guardian Signs effect sheets, item icons and every ball sprite, fonts, five music tracks, UI art, the browser icon (Wesley's pixel Lugia head), and the baked island under assets/maps.
- **Map generator**: scripts/fantasy/*.py (Python with numpy, scipy, Pillow), described in section 7.
- **Tests and checks**: Vitest unit tests (81 on online, 56 on main), server/headless.ts (a whole match without a network), server/checks (social, teams, lagcomp, hitch, moderation, ops, launch_check), server/bots.ts (fake players against a running server), ci.yml on every push.
- Size: about 42,000 lines of TypeScript and Python, 294 commits.

3. Repository layout

Path	What lives there
src/core	fixed-step loop, input, seeded rng, simplex noise, math
src/sprites	AnimData and Offsets parsers, sprite registry and animation player, ranger sheets, item icons, asset loader (missing art draws a magenta labelled box, never a crash)
src/ground	GroundRenderer interface; fanGround.ts bakes the Fan-tasy map in chunks (tile layers, animated tiles, the paving overlay, the colour blend); older procedural, Sunnyside and static renderers remain as fallbacks
src/render	camera and view, world drawing with depth-sorted objects, WebGL lighting
src/sim	the whole game simulation
src/ui	title, starter roll, character screen, match screen and HUD, end screen, PC, trade menu, social panel, online screens (hub, lobby), options, move gallery, watch screens, overlay (the cream panel style)
src/net	protocol, sync, mirror, connection, progression (trainer XP, looks, achievements, titles)
src/vfx, src/audio	effect library and visuals, floating text, music
src/pc.ts, src/chest.ts, src/look.ts, src/settings.ts, src/stats.ts, src/dev.ts	offline persistence in localStorage (PC box, storage chest almia.chest.v2, trainer look, settings almia.settings.v1, stats) and dev flags
server/	hub, matchmaker, room and workers, accounts, chat, moderation and the mod dashboard, rating (Glicko-2), replays, backups and S3, telemetry, events, privacy and profile pages, admin, bots, checks
scripts/fantasy	the island generator: design3, paint3, cliffs, automap, wang, tiled, castlekit, castle3, farm3, content3, decor3, polish3, animprops, areas, towns3, statue, stamps, export3, build3
scripts/mapgen	post-render passes: pave_smooth.py, center_blend.py, minimap_from_render.py
scripts/*.py	data builders: fetch-sprites.py (sparse SpriteCollab checkout and credits), learnsets.py, moves.py, megas.py, shiny-palettes.py, add_starters.py
tools/mapshot.html, src/tools/mapshot.ts	renders the island in the real engine (layer=ground|decor|all, blend=0/1, overlay=0/1, skip=layers)
docs/	online.md, launch-day.md, plan.md, project-brief-2026-10-27.pdf (this document)

4. Gameplay reference

Everything a player meets, with the numbers the code uses (config.json unless noted). P$ stands for Poké Dollars.

4.1 Title screen and starting a match

- The title screen is a live corner of the island: wild Pokémon wander in, Darkrai puts them to sleep with Dark Void and blasts them, now and then Cresselia flies in, wakes them and fights Darkrai off. Buttons: Play (offline), Play Online (O), Trainer (C, the look picker), My PC (P), Move Gallery (M), Demo Team (D), Options.
- **Roll Starter**: three reels each roll a rarity, then a Pokémon of that rarity. Pick one (click or 1-3); Reroll (R) deals a fresh three (unlimited outside a match). A new player rolls from the common and uncommon tiers only; every Pokémon kept in the PC adds the first form of its line to the pool. Each starter comes with its whole evolution line.
- You start at level 5 at a random spot near the coast; seven rival trainers (Rook, Marlo, Wynne, Brigg, Tamsin, Dale and Juniper, Ranger Union members from the Ranger sheets) are spread round the coast away from you, each with a party of three from the trainer pool (Turtwig, Chimchar, Piplup, Pikachu, Eevee, Riolu, Gible, Dratini, Larvitar, Shinx, Geodude, Psyduck). Every trainer carries the starting kit: three Poké Balls and a Potion, and P$500.
- Online: a lobby of up to eight humans; computer trainers fill the rest once at least half the seats are human (MIN_PLAYERS = ceil(size/2)); the lobby shows "Waiting for N more trainers"; Space starts (Enter no longer does); "Start now with bots" works in casual lobbies once the minimum is met.

Rarity	Chance	Pokémon
Common	50%	Pikachu, Meowth, Magikarp, Psyduck, Geodude, Shinx, Starly, Bidoof
Uncommon	27%	Elekid, Magby, Smoochum, Tyrogue, Munchlax, Eevee
Rare	13%	every region's starter trio from Kanto to Paldea (27) and Riolu
Epic	7%	Dratini, Larvitar, Bagon, Beldum, Gible, Deino, Goomy, Jangmo-o, Dreepy, Frigibax
Mythic	3%	Mew, Celebi, Jirachi, Manaphy, Shaymin, Victini, Marshadow, Meltan

4.2 Controls

Input	Action
WASD / arrows	move, 8 directions, 5.5 tiles/s; the party steers around you in formation (leash 4.2 tiles, 2.6 for the lead, 2.0 for a Passive one)
mouse	aim: a reticle marks where a single-target move lands (up to 5.6 tiles); within 1.6 tiles of an enemy it snaps onto it (red)
left / right click, Q / E, 1-2	cast move 1 / move 2 at the aim point (hold to keep casting); 0.3 s cast, telegraph on the ground, cooldowns 1.5-6 s; lines and cones fire toward the aim
click one of your Pokémon	Pokémon mode: you steer it, its moves fire the way it faces at the nearest enemy roughly ahead within 6 tiles; the camera rides with it; Esc or a click on your trainer returns to trainer mode
R	auto-battle on/off (on at the start): the lead picks its own moves and targets while you steer, also in one-on-one battles
Shift+1-6, or click the bag	use an item; a ball's key arms it and again cycles the ball type
G	switch between the Bag and the Held & TMs tab; Shift+1-6 there gives a held item or teaches a TM (pick the move to replace with 1-2, Esc keeps)
Space beside a person	talk: vendor (1-9 buys, Space/Esc leaves), healer, the Castle Champion (challenge), a townsperson (their lines), the Breeder (trades an Egg for her wanted species), the Merchant, Celeste, Hoopa
Space beside an item ball	pick it up (a popup reveals the contents); Space at a chest opens the storage chest
Space with nothing in reach	dodge: the lead dashes 2.2 tiles over 0.18 s the way you move or face; hits in the first 0.3 s miss; 3 s cooldown. At the water's edge with a rod: fish
hold C and draw loops	Capture Styler (needs the Styler held item): loops closed round a wild calm it; see 4.9
F (Tab or click a wild to retarget)	the ball locks onto the nearest catchable wild within 9 tiles (brackets, green in range); Tab steps through the wilds in range; clicking any wild locks it; F throws the armed ball, 7 tiles

Input	Action
M	Mega Evolve the Pokémon you steer (or the lead) when it holds its own Mega Stone; once a match
V	fly on Latios or Flygon, ride Raikou or Torterra, if one is in the party; V again gets off
click a portrait	swap the lead; click a behaviour chip to cycle Attack / Guard / Passive
hover a party card: red X	release that Pokémon (confirm on "Release?"); it walks off calm and can be caught again; not in a battle, never your last one
hover a bag slot: red X	throw the item away; at a shop, click a slot or Shift+1-6 sells one for half the lowest shop price
Shift+number on a Crystal Shard	clears the darkness in a 5-tile circle round you for 10 s
N, Esc, Options	music on/off; menus; Options rebinds every key (W A S D Q E Space Tab F C V M R G N), music volume, camera shake, simple graphics
1-6 or click a portrait in a battle	send out the next Pokémon when one faints

4.3 The island

320 x 224 tiles of 16 px, a new random start spot every match (the island itself is fixed; ?seed=text changes it). The Shadow Crystal stands in the centre circle of a paved oval battlefield on a terrace in an open plain; the plain is blighted (ash-grey grass, dead trees, stumps, logs, skulls, broken tombstones, glowing rune rocks, braziers on the terrace, ruined pillars, purple banners, crystal clusters). Four lands surround it, each with its own wild table, a boss and a town with a healer, three market stalls and a storage chest:

Land	Where	Town	Notable places
Snowfields	north-west	Frostmere (on the foothill plateau)	the ice mountain (five tiers of ice cliffs above the foothill, a trail of stairs with torches, the summit tower with braziers, banners and rune ice, rangers' camps with shelters and campfires, a watchtower, flags), a stream that falls over the foothill face into the frozen lake as a waterfall
Autumn wood	north-east	Emberwood	a ridge of brown cliffs along the coast, a farm, the river from Tide Lake to the north coast with a stone bridge on the Emberwood road, cherry trees
Desert	south-west	Sunreach (inside a low stone wall)	mesas, an oasis with palms and lily pads, a ruined temple (arches, pillars, fallen walls, vases, bones, braziers, rune rocks), a paid Crystal Healer's outpost is on the plain side
Forest	south-east	Greenhollow (inside a low stone wall)	the castle in its clearing, the forest lake, a hill on the coast, a farm, the thickest woods
The plain	centre		the arena, the blight, Tide Lake (Lugia) between the plain and the autumn wood, the Crystal Healer (P$600 a heal) south of the arena

- Roads wind between everything (never one or two tiles wide), meet towns, gates and stairs head on, and converge on the arena at its north, south, east and west; every plateau has stairs where roads climb and extra flights in every long south face (about every 22 tiles); the snow tiers have flights on their west, east and north faces too. Signposts at junctions, lamp posts along the roads, flowers, mushrooms and logs on the verges.
- Every town stands inside its own enclosure (stone wall or fence) open where the roads come in, with houses facing a brick plaza, three stalls with a vendor each, a well, a healer, a storage chest, torches at the plaza corners, trees, a watchtower and props. Farms outside Greenhollow and Emberwood (fields of corn, cabbages, pumpkins, tomatoes and so on, a fence with a gate, farmhouse, haystacks, cart, well, scarecrow). Boats, nets and fish barrels on the beaches, palms on the southern sands.
- **The castle** (south-east forest): a walled hold composed from the pack's castle kit: a plank walkway ring with parapets, round towers on the corners, a gatehouse in the south wall, gaps in the west and east walls with red-roofed towers either side, the keep on the north wall with stairs down into the courtyard. Inside: kerbed lawns, a paved plaza round the battlefield oval, walks from every gate, a fountain with a stone Celebi, cypresses, hedges, flower beds, benches, lamps, a training yard, stores, a camp, a free healer, and the Castle Champion.
- Minimap in the HUD (the painted island, dots for the merchant, Celeste, Hoopa, towns, the storm ring).

4.4 Wild Pokémon: the complete spawn tables

Each land has a fixed table that is always used (every weight jittered 0.6x to 1.4x per match), plus **five species drawn at random from its rotation pool** each match (rotationPicks = 5), so the mix differs every game. Weights: 10 is common, 4 uncommon, 1.2 rare; the table's level range applies to common entries and an entry's own range overrides it. Every Mega-capable line (megaLines) always has a first form placed in its land: Charmander and Gastly in the autumn wood, Snorunt and Absol in the snow, Riolu and Ralts in the forest, Larvitar in the desert. Counts at the start: forest 46, autumn 42, snow 44, desert 38, plus 26 in the meadows and 18 on the shores, which use the nearest land's table (about 218 wilds, 241 forms possible). Levels climb toward the crystal by up to 55 (quadratic over 140 tiles, spread 0.72x to 1.08x), so wilds near the centre are met evolved. Wilds wander within 4 tiles at 2.2 tiles/s, aggro at 5.5 tiles, bosses at 4.

Land (levels)	Fixed table: species (weight, own levels)
Forest (2-5)	Grubbin (10), Starly (10), Applin (8), Budew (10), Wurmple (12), Tropius (4, Lv 5-8), Turtwig (1.2, Lv 5-7), Treecko (1.2, Lv 5-7), Ralts (1.2, Lv 5-7), Munchlax (1.2)
Autumn wood (3-6)	Solosis (10), Fomantis (10), Impidimp (10), Charcadet (8), Deerling (10), Musharna (4, Lv 8-11), Zorua (1.2, Lv 7-10)
Snowfields (3-6)	Snover (10), Alolan Vulpix (10), Amaura (7), Spheal (10), Snom (10), Cubchoo (8), Absol (4, Lv 7-10), Sneasel (4, Lv 6-9), Frigibax (1.2, Lv 6-9)
Desert (4-7)	Sandile (10), Trapinch (10), Darumaka (10), Hippopotas (5), Larvitar (1.2, Lv 8-11)

Rotation pool (5 drawn per match)	Species (weight)
Forest, 26	Caterpie (4), Kricketot (4), Grubbin (4), Wurmple (4), Budew (4), Fomantis (4), Starly (4), Bidoof (4), Hoothoot (4), Sentret (4), Pikachu (4), Shinx (4), Applin (4), Tropius (4), Eevee (1.2), Tyrogue (1.2), Goomy (1.2), Bulbasaur (1.2), Chikorita (1.2), Treecko (1.2), Turtwig (1.2), Snivy (1.2), Chespin (1.2), Rowlet (1.2), Grookey (1.2), Sprigatito (1.2)
Autumn wood, 23	Vulpix (4), Charcadet (4), Zorua (4), Impidimp (4), Teddiursa (4), Deerling (4), Meowth (4), Munchlax (4), Solosis (4), Musharna (4), Honedge (4), Chimchar (1.2), Magby (1.2), Elekid (1.2), Dreepy (1.2), Deino (1.2), Cyndaquil (1.2), Torchic (1.2), Tepig (1.2), Fennekin (1.2), Litten (1.2), Scorbunny (1.2), Fuecoco (1.2)
Snowfields, 16	Swinub (4), Spheal (4), Amaura (4), Bronzor (4), Bergmite (4), Smoochum (1.2), Frigibax (1.2), Beldum (1.2), Dratini (1.2), Piplup (1.2), Froakie (1.2), Popplio (1.2), Sobble (1.2), Quaxly (1.2), Totodile (1.2), Squirtle (1.2)
Desert, 10	Sandshrew (4), Hippopotas (4), Geodude (4), Cacnea (4), Darumaka (4), Bronzor (4), Gible (1.2), Bagon (1.2), Jangmo-o (1.2), Mudkip (1.2)

Bosses (one per land, deep inside it, 3x XP)	Ursaring Lv 20 (forest), Gallade Lv 26 (autumn wood), Avalugg Lv 28 (snowfields), Steelix Lv 36 (desert)
Rival trainers' pool (3 each)	Turtwig, Chimchar, Piplup, Pikachu, Eevee, Riolu, Gible, Dratini, Larvitar, Shinx, Geodude, Psyduck
Starter pool for a PC-less player	the Common and Uncommon tiers of 4.1; the full pool (60) is those plus the Rare, Epic and Mythic tiers, unlocked line by line through the PC

- **Elite** (12% of spawns): 5-9 levels higher, often already evolved, a gold name and star, 1.6x XP.
- **Alpha** (7% of spawns within 55 tiles of the crystal): about 10 levels above the wilds around it (at most 65), always the last form of its line, drawn 1.5x bigger with a red glow and embers, 2.5x XP. Knocking one out or catching it always drops an Ultra Ball prize: a rare TM (40%), two Rare Candies (35%) or a rare-tier roll (25%), plus 12% for an Extraction Voucher; an Alpha of a Mega line drops its Mega Stone 30% of the time.
- **Shiny** (0.2% of spawns, four times that for Alphas): real SpriteCollab shiny colours as palette swaps, a light-blue "Shiny" name and a glint; stays shiny when caught and when it evolves.
- Wild knockouts drop an item ball 40% of the time (rolled from the loot tiers). 120 item balls lie on the ground at the start. Respawning wilds: the land tables keep their counts topped up through the match.

4.5 Fighting

- Every trainer has at most **two Pokémon out** (the first two in party order that can fight); the others wait in their balls and come out when one faints or on a swap. The party cap is 6 (5 for the demo team).

- A Pokémon knows **two moves**: the newest two attacks reached in its learnset (a status move only fills a slot no attack can). On a level-up a free slot takes the new move; with both full a stronger new move replaces the weaker one. TMs ask which to forget.
- Casting: 0.3 s cast, a telegraph on the ground, cooldowns 1.5-6 s, range scaled 1.6x. **Channelled beams** (Flamethrower, Ice Beam, Thunderbolt, Dragon Pulse, Hydro Pump, Solar Beam, Flash Cannon, Aeroblast, Psybeam, Aurora Beam, Signal Beam, Dark Pulse) keep firing for 2 s with a pulse every 0.25 s, 1.3x the move's power spread over the pulses, steered by the cursor (turning at most 140 degrees a second); the user stands still. **Lunges** (contact moves: Tackle, Bite, Close Combat...): the user rushes at 16 tiles/s for at most 0.45 s and strikes on arrival; a contact hit shoves the target 0.45 tiles (times power/80) unless it is a boss, legend, Alpha or much bigger. Body Slam and Heavy Slam are leaps with a shockwave on landing.
- Damage: the type chart (18 types), STAB 1.5x, crits 6%, every hit scaled by 0.55 so fights last several exchanges, 60 ms hit-stop. Movement speed follows the Speed stat (slow and fast bases mapped to a multiplier range), Speed stages from Agility, paralysis slows.
- Statuses: burn (half physical damage, ticks 6.25%), paralysis (slower, sometimes skips a move), sleep, poison (7% of max HP every 2 s until cured; Poison and Steel immune), freeze (2.5-4 s, a Fire hit thaws; Ice and Fire immune), confusion (4-6 s, 40% of moves go off in a random direction). Poison, ice and psychic moves cause them. Healing at an outpost cures all.
- Behaviour chips per Pokémon: Attack (goes after anything within 12 tiles), Guard (fights what comes within 5), Passive (never starts a fight, dodges, stays close). Auto-battle picks moves and targets for the lead.
- Held items: Leftovers, Focus Sash, Choice Band, Life Orb, Exp. Share, Lucky Egg, the fourteen type boosters (Charcoal, Mystic Water, Magnet, Miracle Seed, NeverMeltIce, Black Belt, Soft Sand, Sharp Beak, TwistedSpoon, Poison Barb, Hard Stone, Spell Tag, Dragon Fang, BlackGlasses, Metal Coat, Silk Scarf), the ten Mega Stones. Charms work from the bag: Storm Mask, Capture Styler, Snag Machine, the Breeder's Egg.
- Moves: 175 in the library (the complete table follows), each with one animation recipe from the Guardian Signs sheets (a carrier, one impact, at most one aftershock). Shapes: single (aimed at one target), line (a beam or bolt along the aim), cone (a spread in front), ring (around the user), self (a buff). Signature moves always known by their legend: Aeroblast (Lugia), Sacred Fire (Ho-Oh), Dragon Ascent (Rayquaza), Luster Purge (Latios), Doom Desire (Jirachi), Hyperspace Hole (Hoopa), Spectral Thief (Marshadow), V-create (Victini), Dark Void and Shadow Storm (Darkrai and the shadow legends). The Move Gallery on the title screen casts every move in turn at a dummy.

All 175 moves (power / shape / cooldown in seconds; status moves have no power)

Type (count)	Move power / shape / cooldown
Bug (7)	X-Scissor 80/single/3.5s, Bug Bite 60/single/2.75s, String Shot status/cone/5s, Struggle Bug 50/cone/3.0s, Fury Cutter 40/single/2.0s, Signal Beam 75/line/5.0s, Bug Buzz 90/ring/5.25s
Dark (10)	Bite 60/single/2.75s, Dark Pulse 80/line/5.0s, Dark Void status/ring/8s, Crunch 80/single/3.5s, Feint Attack 60/single/2.75s, Night Slash 70/single/3.25s, Snarl 55/cone/3.25s, Sucker Punch 70/single/2.4s, Nasty Plot status/self/8s, Shadow Storm 75/ring/6s
Dragon (8)	Dragon Pulse 85/line/5.25s, Dragon Breath 60/cone/3.75s, Dragon Claw 80/single/3.5s, Twister 40/line/2.5s, Outrage 120/single/4.75s, Draco Meteor 130/single/4.5s, Dragon Dance status/self/8s, Dragon Tail 60/single/2.75s
Electric (11)	Thunder Shock 40/line/2.5s, Discharge 80/ring/5.25s, Thunder Wave status/single/5s, Thunderbolt 90/line/5.75s, Wild Charge 90/single/3.5s, Spark 65/single/3.25s, Thunder Fang 65/single/3.0s, Thunder Punch 75/single/3.5s, Charge Beam 50/line/3.0s, Thunder 110/single/5.0s, Nuzzle 20/single/4.0s
Fairy (7)	Fairy Wind 40/single/2.0s, Dazzling Gleam 80/ring/4.75s, Disarming Voice 40/ring/2.75s, Draining Kiss 50/single/2.5s, Moonblast 95/single/4.25s, Play Rough 90/single/4.0s, Charm status/single/6s
Fighting (10)	Rock Smash 40/single/2.25s, Brick Break 75/single/3.25s, Karate Chop 50/single/2.5s, Low Kick 60/single/2.75s, Double Kick 60/single/2.75s, Force Palm 60/single/3.0s, Aura Sphere 80/single/3.5s, Close Combat 120/single/4.25s, Bulk Up status/self/8s, Mach Punch 40/single/1.75s
Fire (13)	Ember 40/single/2.0s, Will-O-Wisp status/single/5s, Flamethrower 90/line/5.75s, Flame Wheel 60/single/3.0s, Fire Fang 65/single/3.0s, Fire Punch 75/single/3.5s, Fire Spin 35/single/2.0s, Heat Wave 95/cone/4.75s, Fire Blast 110/single/4.5s, Sacred Fire 100/single/6s, V-create 180/single/10s, Flame Charge 50/single/3.0s, Flare Blitz 120/single/4.25s
Flying (8)	Wing Attack 60/single/2.75s, Aerial Ace 60/single/2.75s, Gust 40/single/2.0s, Peck 35/single/1.75s, Air Slash 75/single/3.25s, Brave Bird 120/single/4.25s, Aeroblast 100/line/6s, Dragon Ascent 120/single/8s

Type (count)	Move power / shape / cooldown
Ghost (8)	Shadow Ball 80/single/3.5s, Lick 30/single/1.75s, Astonish 30/single/1.5s, Shadow Claw 70/single/3.25s, Hex 65/single/3.0s, Night Shade 50/single/2.5s, Shadow Sneak 40/single/1.75s, Spectral Thief 90/single/5.5s
Grass (13)	Razor Leaf 55/cone/3.0s, Sleep Powder status/cone/6s, Energy Ball 80/single/3.5s, Leafage 40/single/2.0s, Stun Spore status/cone/6s, Vine Whip 45/single/2.25s, Absorb 30/single/1.5s, Mega Drain 40/single/2.0s, Giga Drain 75/single/3.25s, Magical Leaf 60/single/2.75s, Seed Bomb 80/single/3.5s, Leaf Blade 90/single/3.75s, Solar Beam 120/line/6.75s
Ground (8)	Bulldoze 60/ring/3.75s, Earthquake 100/ring/5.5s, Mud Shot 55/single/2.75s, Mud-Slap 30/single/1.5s, Sand Attack status/cone/5s, Sand Tomb 35/single/1.75s, Drill Run 80/single/3.5s, Earth Power 90/single/3.75s
Ice (10)	Ice Beam 90/line/5.75s, Powder Snow 40/cone/2.5s, Ice Shard 40/single/1.75s, Icicle Crash 85/single/3.75s, Ice Fang 65/single/3.0s, Ice Punch 75/single/3.5s, Aurora Beam 65/line/4.25s, Icy Wind 55/cone/3.25s, Avalanche 60/single/2.75s, Blizzard 110/ring/6.25s
Normal (17)	Tackle 40/single/2.0s, Scratch 40/single/2.0s, Quick Attack 40/single/1.75s, Howl status/self/6s, Return 90/single/3.75s, Growl status/cone/5s, Leer status/cone/5s, Scary Face status/single/6s, Headbutt 70/single/3.25s, Body Slam 85/single/4.0s, Double-Edge 120/single/4.25s, Hyper Voice 90/ring/5.25s, Swift 60/single/2.75s, Slash 70/single/3.25s, Sing status/single/7s, Harden status/self/6s, Swords Dance status/self/8s
Poison (7)	Poison Jab 80/single/4.0s, Sludge Bomb 90/single/4.25s, Poison Sting 25/single/1.5s, Acid 40/cone/2.5s, Smog 30/cone/2.0s, Poison Fang 50/single/2.75s, Sludge 65/single/3.25s
Psychic (10)	Psybeam 65/line/4.5s, Confusion 50/single/2.5s, Psycho Cut 70/single/3.25s, Hypnosis status/single/6s, Agility status/self/8s, Psychic 90/single/4.0s, Zen Headbutt 80/single/3.5s, Calm Mind status/self/8s, Luster Purge 95/single/6s, Hyperspace Hole 80/single/5s
Rock (7)	Rock Throw 50/single/2.5s, Rock Slide 75/cone/4.0s, Stone Edge 100/single/4.25s, Rock Tomb 60/single/3.0s, Ancient Power 60/single/3.0s, Smack Down 50/single/2.5s, Power Gem 80/single/3.5s
Steel (10)	Iron Head 80/single/3.5s, Metal Claw 50/single/2.5s, Iron Tail 100/single/4.25s, Flash Cannon 80/line/5.0s, Bullet Punch 40/single/1.75s, Steel Wing 70/single/3.25s, Iron Defense status/self/8s, Heavy Slam 80/single/3.5s, Doom Desire 140/single/8.5s, Mirror Shot 65/single/3.25s
Water (11)	Water Gun 40/single/2.0s, Surf 90/ring/5.25s, Bubble 40/cone/2.5s, Bubble Beam 65/single/3.0s, Water Pulse 60/single/3.0s, Aqua Tail 90/single/3.75s, Hydro Pump 110/line/6.25s, Aqua Jet 40/single/1.75s, Razor Shell 75/single/3.75s, Waterfall 80/single/3.5s, Wave Crash 120/single/4.25s

4.6 One-on-one trainer battles

- Two trainers within 6 tiles of each other in sight start a battle: both parties go back into their balls, each sends out one Pokémon 1.6 tiles in front, the trainers step 9 tiles apart, the fight stays within 11 tiles of the middle, the camera zooms 1.15x and your trainer stands back. You steer your Pokémon (4.2 tiles/s) and cast its moves the way it faces (straight at the opponent when roughly ahead); when it faints, pick the next (1-6 or click). A rival picks its next after 1.2 s and dodges telegraphed moves 35% of the time. The loser is out of the match; the winner's team is not healed. 12 s before the same pair can fight again. Squad teammates never fight each other.
- The **Castle Champion** waits in the castle courtyard (Space beside him): Lucario, Garchomp and Gyarados three levels above your strongest (at least 10). Beat his whole team for P$3000, three Ultra Balls, two Hyper Potions, a Life Orb and a shiny Pokémon at your best level; a loss is friendly (he withdraws, your Pokémon are patched up at the castle healer, try again after 60 s). Once per match.
- **Snag Machine**: Shift+number beside a rival's fainted Pokémon claims it (wear it down under 10% first); it also takes Shadow Lugia under 10% of its HP.
- Online: a player who drops mid-battle keeps fighting (the lead picks its own moves) until the 45 s reconnect window runs out.

4.7 Levels, XP and evolution

- XP to the next level = 30 + 5L + 0.15L^2. A knockout pays 27 + 10.8 per level of the fallen Pokémon (up to double against higher levels); the partner out beside the dealer takes half, Pokémon in their balls 40% (Exp. Share: full), a catch gives the lead 75% of a knockout. Elite 1.6x, boss 3x, Alpha 2.5x, Lucky Egg 1.5x. An even fight is worth about a level and a quarter early on, one level around 25, under a level by 40.
- Evolution at canonical levels (species.json evolves; Eevee picks at random), a glow and a swap in the field; wilds spawned above an evolution level appear evolved. Stones (Fire, Water, Thunder, Leaf, Ice, Moon, Sun, Shiny, Dusk, Dawn) evolve the lead when its species has that stone evolution (Pikachu, Eevee's stone forms, Kirlia to

Gallade, Roselia, Doublade, Charjabug, Vulpix, Alolan Vulpix); each town's medicine stall sells its own stone and the Merchant has a stone of the day; stones are never in item balls. Rare Candy gives the lead a level.

- Rivals hunt wilds at most 3 levels above their team within 30 tiles while above half HP, go back to an outpost within 70 tiles to heal, and roam now and then; they Mega Evolve as soon as a stone holder is fighting and flee under 30% HP.
- **Mega Evolution**: give a Pokémon its own Mega Stone, press M, and it becomes the Mega form for the rest of the match with the Mega's types and stats; once per trainer per match. Mega Stones come from Great and Ultra Ball loot (usually for a Pokémon you have) and Alpha drops (30%). A Mega sent to the PC is stored as its regular form. The ten Megas (types; HP/Atk/Def/SpA/SpD/Spe): Mega Charizard X (Fire/Dragon; 78/130/111/130/85/100), Mega Gengar (Ghost/Poison; 60/65/80/170/95/130), Mega Steelix (Steel/Ground; 75/125/230/55/95/30), Mega Tyranitar (Rock/Dark; 100/164/150/95/120/71), Mega Gardevoir (Psychic/Fairy; 68/85/65/165/135/100), Mega Absol (Dark; 65/150/60/115/60/115), Mega Glalie (Ice; 80/120/80/120/80/100), Mega Latios (Dragon/Psychic; 80/130/100/160/120/110), Mega Lucario (Fighting/Steel; 70/145/88/140/70/112), Mega Gallade (Psychic/Fighting; 68/165/95/65/115/110).

4.8 Items: all 64, with what each does

Bag: 6 slots, each stacking one item. Items are used with Shift+1-6 or by clicking the bag; held items and TMs are given from the Held & TMs tab (G).

Kind	Items and effects
Balls (7)	Poké Ball x1; Great Ball x1.5; Ultra Ball x2; Master Ball never fails (appears beside the crystal when the final darkness begins; the Merchant sells it); Net Ball x3 on Bug and Water; Dusk Ball x3 inside the darkness; Quick Ball x5 if thrown before the target has been attacked. Status bonus: sleep x2, paralysis and burn x1.5. Throw range 7 tiles, 1.2 s aim, 0.55 s wobble. Placed by the match only: Premier Ball (an eliminated trainer's bag), Cherish Ball (Darkrai's drop: a Dark-type held item and a full party heal).
Medicine (5)	Potion heals 20 HP; Super Potion 50; Hyper Potion 120; Revive brings a fainted Pokémon back at half HP; Max Revive at full.
Battle held items (6)	Leftovers: 1/16 of max HP every 3 s. Focus Sash: survives one hit from full HP with 1 HP. Choice Band: physical moves x1.5. Life Orb: all moves x1.3, 10% of max HP recoil once per move used. Exp. Share: full XP while waiting in its ball. Lucky Egg: 1.5x XP.
Type boosters (16)	x1.2 on moves of the type: Charcoal (Fire), Mystic Water (Water), Magnet (Electric), Miracle Seed (Grass), NeverMeltIce (Ice), Black Belt (Fighting), Soft Sand (Ground), Sharp Beak (Flying), TwistedSpoon (Psychic), Poison Barb (Poison), Hard Stone (Rock), Spell Tag (Ghost), Dragon Fang (Dragon), BlackGlasses (Dark), Metal Coat (Steel), Silk Scarf (Normal).
Mega Stones (10)	Charizardite X, Gengarite, Steelixite, Tyranitarite, Gardevoirite, Absolite, Glalitite, Latiosite, Lucarionite, Galladite: held by the matching Pokémon, M Mega Evolves it once a match.
Evolution stones (10)	Fire, Water, Thunder, Leaf, Ice, Moon, Sun, Shiny, Dusk, Dawn Stone: evolve the lead when its species has that stone evolution. Sold by town medicine stalls (one each) and the Merchant (stone of the day); never in item balls.
Charms and tools (9)	Storm Mask (works from the bag: Pokémon last 2 minutes in the storm, the trainer 4; Merchant only, 5% of matches, P$20000). Capture Styler (hold C and draw loops to calm a wild; rare loot). Snag Machine (takes a rival's fainted Pokémon, or Shadow Lugia under 10%). Crystal Shard (clears the darkness in a 5-tile circle for 10 s). Rare Candy (the lead gains a level). Extraction Voucher (sends one party Pokémon to the PC for keeps). Old Rod, Good Rod, Super Rod (fishing, see 4.10). Pokémon Egg (from Breeder Maribel; hatches after 150 s into Dratini, Larvitar, Bagon, Beldum, Gible, Deino, Goomy, Frigibax, Dreepy or Jangmo-o).
TMs (38)	Common (21, Great Ball loot and stalls): Energy Ball, Brick Break, Poison Jab, Rock Slide, Dragon Pulse, Fire Punch, Thunder Punch, Ice Punch, Shadow Ball, Play Rough, Waterfall, Sucker Punch, Air Slash, Bulk Up, Calm Mind, Aura Sphere, Power Gem, Dark Pulse, Leaf Blade, Will-O-Wisp, Thunder Wave. Rare (17, Ultra Ball loot and Alpha prizes only): Flamethrower, Surf, Thunderbolt, Ice Beam, Psychic, Earthquake, Sludge Bomb, Stone Edge, Wild Charge, Heat Wave, Close Combat, Dragon Dance, Earth Power, Moonblast, Brave Bird, Outrage, Hypnosis. A TM teaches the lead an attack of the kind it fights with (physical, special, or either for a mixed attacker), whatever its type; Bulk Up and Dragon Dance go to physical fighters, Calm Mind to special ones, Will-O-Wisp, Thunder Wave and Hypnosis to anyone; you pick which move to forget.

Item balls on the ground (120 at the start) and 40% of wild knockouts roll a tier, then an entry by weight:

Tier (chance, sprite)	Entries (weight)
Common (62%, Poké Ball)	Poké Ball (16), Potion (12), Super Potion (6), Old Rod (2)
Uncommon (28%, Great Ball)	Great Ball (10), a common TM (6), Hyper Potion (5), Revive (4), Rare Candy (3), Snag Machine (2), Good Rod (2), Net Ball (1), Dusk Ball (1), Quick Ball (1), a Mega Stone (1, usually for a Pokémon in the finder's party), Extraction Voucher (1), Max Revive (1)
Rare (10%, Ultra Ball)	Ultra Ball (8), a rare TM (5), two Rare Candies (4), Max Revive (3), Capture Styler (2), Snag Machine (2), Crystal Shard (2), Exp. Share (2), Lucky Egg (2), Extraction Voucher (2), Super Rod (2), and at weight 1 each: Leftovers, Focus Sash, Choice Band, Life Orb, every one of the 16 type boosters, Net Ball, Dusk Ball, Quick Ball, a Mega Stone

Shops and money

- Start with P$500. A wild knockout pays P$20 + P$3 per level; three times that for a trainer's Pokémon. The Castle Champion pays P$3000.
- Selling: at any shop, click a bag slot or Shift+1-6 to sell one for half the lowest price any stall charges; items no stall sells go by kind (ball 300, medicine 300, revive 1200, held item 2000, common TM 1500, rare TM 3000, Mega Stone 4000, Styler 3000, shard 1500, candy 1200, stone 1800, voucher 6000, rod 1500, halved; never under P$10).
- Every town has three stalls (Space beside the vendor, 1-9 buys). **Poké Ball stall**: Poké Ball 100, Great Ball 300, Ultra Ball 700, Net Ball 400, Dusk Ball 500, Quick Ball 600, Old Rod 300, Good Rod 1500. **Medicine stall**: Potion 150, Super Potion 400, Hyper Potion 900, Revive 1200, Rare Candy 1200, and the town's evolution stone. **Battle gear stall**: Leftovers 2500, Focus Sash 2000, Life Orb 3000, Exp. Share 2500, Lucky Egg 2500, and the town's type item at 1500: NeverMeltIce in Frostmere, Charcoal in Emberwood, Soft Sand in Sunreach, Miracle Seed in Greenhollow, Metal Coat at the castle.
- **Wandering Merchant** (gold dot on the minimap): Rare Candy 900, Super Rod 4000, Master Ball 8000, Extraction Voucher 6000, Leftovers 2200, Life Orb 2600, Crystal Shard 1500, Max Revive 2500, the stone of the day, and in 5% of matches the Storm Mask at 20000.
- **Storage chests**: one beside every town healer, all opening the same store per trainer: 10 slots, each stacking any number of one item. Items in the chest survive a wipe-out and persist between matches (server-side for accounts, localStorage offline). The former dev "rare-item chest" is off unless ?rare=1 or a private lobby turns it on.

4.9 Catching and the Capture Styler

- F locks the nearest catchable wild within 9 tiles (one fighting your team counts as closer), brackets turn green within a throw; a picked target holds until it faints, is caught or gets about 24 tiles away. A worn-down Pokémon is easier; inside the darkness wilds get a purple aura and cannot be caught.
- The **Capture Styler** is a rare held item (Ultra Ball loot) with a spinning cursor icon while in use: hold C and draw loops round a wild within 7 tiles. It needs 2 loops + 1 per 10 levels (+2 for Elites, +4 for bosses, at most 8); a full set calms it for 12 s: it stops fighting and every ball counts 4x. If the Pokémon runs through the line, the line breaks and a loop is lost.
- Legendaries in a bout can be caught at half the ball's usual rate, or join you when worn to their last HP.

4.10 Fishing, flying and riding

- Stand at the water's edge with a rod (Space or use it from the bag): the line goes out 3 tiles (0.6 s cast), a bite comes after the rod's wait, the ! gives 1.3 s to press Space, reeling takes 0.7 s; moving reels in. **Old Rod**: 80% common, 17% uncommon, 3% rare, wait 3-7 s, levels 4-10, items from the common tier. **Good Rod**: 55% / 30% / 10%, 5% an item ball (uncommon tier), wait 2.5-6 s, levels 12-22. **Super Rod**: 30% / 35% / 22%, 10% an item ball (rare tier), 3% the legend (Lugia, Tide Lake only, once a match; elsewhere the roll counts as rare), wait 2-5 s, levels 22-32. All plus 80% of the local wild level bonus, met evolved. A hooked Pokémon bursts out and fights you.
- Fishing pools (weight): **common** Magikarp (30), Goldeen (14), Tentacool (12), Remoraid (10), Finneon (10), Barboach (8), Luvdisc (8), Horsea (8); **uncommon** Chinchou (12), Wailmer (10), Seaking (8), Octillery (8), Lumineon (8), Whiscash (8), Qwilfish (8), Corsola (8), Tentacruel (6), Seadra (6); **rare** Gyarados (14), Lapras (12), Dratini (10), Feebas (10), Relicanth (8), Kingdra (4), Milotic (4), Wailord (4).

- Mounts (V): Latios and Flygon fly (2.1x walking speed, 1.2 tiles up, over water, out of trainer battles); Raikou and Torterra are ridden (1.7x). The mount sits out of fights while carrying you; no landing on water, fishing or steering a Pokémon while mounted; a battle sets you down.

4.11 Visitors, legends and the Shadow Crystal

- **Wandering Merchant** (gold dot): Rare Candy 900, Super Rod 4000, Master Ball 8000, Extraction Voucher 6000, Leftovers 2200, Life Orb 2600, Crystal Shard 1500, Max Revive 2500, the stone of the day, and in 5% of matches a Storm Mask at 20000 (your Pokémon last 2 minutes in the storm instead of 4 s; the trainer 4 minutes).
- **Ace Trainer Celeste** (pink dot): a one-on-one battle with Garchomp, Lucario and Gyarados a few levels above your best. Win: a shiny Lucario, Garchomp, Dragonite, Tyranitar or Gyarados, or a powerful Pokémon (Metagross, Salamence, Goodra, Dragapult, Dragonite, Tyranitar, Koraidon, Miraidon) at your level (an Extraction Voucher if the party is full). Lose: she patches your team; again a minute later.
- **Hoopa** (violet dot, in about 15% of matches; ?hoopa=1 forces it): Space opens a ring portal; a legend (Ho-Oh, Suicune, Latios, shiny Rayquaza, Raikou, Koraidon, Miraidon, Zacian, Zamazenta) steps out at your best level + 20 (at least 60, at most 100) with triple HP and challenges you to a legend bout. Asked inside the storm, Hoopa reaches into the dark instead and **Shadow Mewtwo** comes through. Hoopa enrages after three hits from a Pokémon you steer (Hoopa Unbound). Once per match.
- **Lugia** sleeps in Tide Lake: a Super Rod cast there hooks it 3% of bites, once a match; it bursts out in a column of spray, leaps ashore and fights as a legend. Once the lake is inside the darkness the cast draws **Shadow Lugia** instead (level 100, triple HP, Aeroblast beam for 3 s, Shadow Storm rings every few seconds, balls bounce off; a Snag Machine takes it under 10% HP; it vanishes if it wins or is worn to nothing).
- **Legend bouts** are fought like trainer battles: your team goes back in its balls, one Pokémon at a time. A legend keeps to mid range and strafes, dashes out of telegraphs, recovers moves faster as it weakens, follows one cast with another under two thirds HP, and is enraged (Attack and Speed up) under a third. Worn to its last HP it joins you (or leaves a Voucher with a full party); if it beats your whole team it vanishes and your Pokémon are patched up; it never knocks you out of the match. If you are already battling when it appears it stays out as a wild boss.
- **Townsfolk**: 7-10 a match from a pool of 16 wander the towns and talk (Space cycles their lines). **Breeder Maribel** appears in 35% of matches wanting one common species (Pikachu, Eevee, Magikarp, Gastly, Turtwig, Chimchar, Piplup or Riolu) in exchange for a Pokémon Egg. Every townsperson and every line:

Townsperson	Lines
Youngster Cole	My Bidoof could beat your whole team! ...Okay, maybe not. But he tries really hard! / Keep two Pokémon out and the second one watches your back. I learned that the hard way. / Shorts are comfy and all, but nobody warns you about running from an Alpha in them.
Old Fisherman Walt	Sixty years I fished these waters. The lake out east still gives me the shivers. / Some nights the black water goes still, like it's holding its breath. Don't ask me what for. / A Super Rod isn't about the fish you can catch. It's about the fish you shouldn't.
Hiker Bruno	I hiked all five lands in one day! My knees would like a word with whoever built those stairs. / Snow, sand, leaves, grass: every Pokémon has its ground. Hunt where your target lives.
Lass Marina	I counted the ducks on the summit lake. It is never the same number twice. NEVER. / A boy told me golden rings open in the sky when nobody looks. He also eats sand, so who knows.
Instructor Hale	A worn-down Pokémon goes calm sooner or later. That calm moment is when a ball truly counts. / Auto-battle is fine for walking. For a real fight, take the controls yourself. R, trainer. R. / Your party heals free at any outpost. Dying because you were too proud to walk there is a choice.
Gravekeeper Morrow	These graves are older than the Union. Older than the castle. Mind your step after dark. / When the storm swallows the island, I see shapes walking in it that have no owner.
Scholar Imani	The Shadow Crystal holds the night at bay. One old text calls it 'the lid on a bad dream'. / Crystal shards push the darkness back for a moment. Cheap insurance, if you ask me.
Chef Porcini	I came here to open a restaurant. My only customers are Pidgey. They do not tip. / You'd think a battle royale would be good for business. Everyone is too busy to eat!
Gambler Dice	Every storage chest on this island remembers you between matches. I keep my winnings in one. And my losses. / One in five hundred. Those are shiny odds. I've beaten worse, friend. I've also lost to better.

Townsperson	Lines
Veteran Sarge	Snag Machine rule one: wear the dark ones down under ten percent, THEN fire. Rule two: see rule one. / I was at the first showdown. Back then the storm was just weather. Now it hunts.
Poké Maniac Gus	I collect Pokémon shaped like balls. Voltorb, Jigglypuff, Marill... my collection ROLLS. / An Electrode beat me at hide and seek once. Round object. Flat field. Explain THAT.
Swimmer Lena	They say Latios will carry a trainer he trusts. The sky, the spray, the speed... imagine it. / I swim everywhere. Piers are for people who fear fun.
Blacksmith Orin	I shape quarry ore into charms. The crystal ones hum at night. I stopped asking why. / Evolution stones turn up in the town stalls. Each town stocks its own, so check them all.
Mystic Vela	I read fortunes. Yours says 'beware the final two'. That will be 200 Poké Dollars. / The darkness does not close in to kill you, dear. It closes in to WAKE something.
Tourist Chad	Back home we don't have giant storms or ghost whales. We have a mall. This is better. / I asked a Ranger where the gift shop is. Still waiting on that answer.
Picnicker June	I set out six sandwiches and turned my back for ONE second. Six Pidgey. Zero sandwiches. / Stock potions before you pick fights. The healer cannot reach you mid-battle, friend.

- **The darkness**: every 200 s the Shadow Crystal pulses (five pulses: safe radii 140, 100, 65, 35, 12 tiles) and the darkness creeps in at 0.7 tiles/s. In the dark a Pokémon is drowsy after 4 s and falls asleep, sleepers take Bad Dreams damage (6.25% a second), wilds cannot be caught; 3 s back in the light wakes them; trainers last twice as long as their Pokémon then collapse (wiped out, or out of the match once the final showdown has started). Daylight shifts cold over 13 minutes. When the last circle closes, Darkrai appears at the crystal at level 100 and hunts the remaining trainers with Dark Void rings and Dark Pulse lines. Camera shake happens only for Lugia, Darkrai and Shadow Mewtwo (and the crystal shattering).
- **Final showdown**: when two trainers remain they are called to the arena after a 5 s warning, placed at either end with teams restored (a team far behind is lifted and evolved to three levels under the strongest finalist), and battle one-on-one for the match. The switch duel.finalShowdown in config.json is currently false, which lets the last two find each other instead; the arena remains the centre of the last safe circle either way.

4.12 Healing, respawning, elimination

- Healers: one in each town and one in the castle courtyard, free, a full party heal every 20 s (3 s to heal, 1.8 tile reach). The Crystal Healer south of the arena charges P$600 and closes during the final.
- If wild Pokémon knock out your whole party you are not out: every item drops where you fell (one Premier Ball) and you come back at the nearest outpost outside the darkness, party healed, after 8 s + 4 s per minute played (at most 90 s). Losing to another trainer or to the darkness knocks you out; nobody respawns once the final showdown has started. Rivals follow the same rules. Eliminated players can watch the rest of the match (their squad first).

4.13 Between matches

- **The PC**: win a match and pick one Pokémon from your party on the victory screen, or use an Extraction Voucher mid-match (not during a battle, never your last fighter): that Pokémon leaves in a beam of light. My PC shows everything kept, with shinies, Alphas, shadows and legends marked, sprites sized and centred with a hover animation; a PC-kept line adds its first form to your starter pool. Offline the PC lives in localStorage; online on the server.
- **Trading** (accounts only): look a trainer up by name, offer one of yours for one of theirs; asking back accepts; the server swaps. **Wonder Trade**: put a Pokémon in the pool and get back whatever the next player put in (news in the trade menu). Clicking a name in chat opens their trainer card (Wins, Eliminations, Matches, In PC) with a Trade button.
- **Global chat** on the right of the hub (signed-in players, 200 characters, 5 per 10 s, filtered). Friends by trainer name, parties of up to four playing as a squad, notices, leaderboards, replays, private lobbies by five-letter code.
- **Stats** kept per account: wins, eliminations, matches, catches, knockouts, damage, time survived, and the achievements below.

4.14 Progression, looks, achievements

- Trainer XP is earned in every casual or ranked match (playing, each trainer outlasted, winning, knockouts, catches, damage, time survived; ranked +25%) and listed on the end screen; trainer levels and the level-gated Ranger looks exist in code but are switched off for now (LEVELS_ON = false), so the first Ranger is free and all looks are available offline; the townsfolk looks stay locked.
- Titles come from achievements and are shown on profiles and under names in battles.

Achievement	Condition	Title
First Steps	finish an online match	
Last One Standing	win a match	Champion
Seasoned	win 10	Veteran
Island Legend	win 50	Legend of Almia
Stronger Together	win as a squad	Squad Leader
Podium	top three in a field of four or more	
Unstoppable	10 knockouts in one match	Unstoppable
Gotta Catch Them	5 catches in one match	
Collector	100 catches	Collector
Heavy Hitter	5,000 damage in one match	Heavy Hitter
Storm Chaser	survive 10 minutes	
Full House	finish with six Pokémon	
Something Sparkly	finish with a shiny	Shiny Hunter
Alpha Wrangler	finish with an Alpha	
Legend Tamer	finish with a legendary	Legend Tamer
Raised Right	finish with a level 50+ Pokémon	
Ranked Victor	win a ranked match	
Gold Standard	reach Gold in ranked	
Diamond Hands	reach Diamond in ranked	Diamond
Master Ranger	reach Master in ranked	Master Ranger
Starting a Collection	keep 10 Pokémon in the PC	
Professor	keep 50 Pokémon in the PC	Professor
Ace Trainer	reach trainer level 10	Ace Trainer
Elite	reach trainer level 25	Elite
Almia's Finest	reach trainer level 50	Almia's Finest

4.15 Audio, options and URL flags

- Music: five looping tracks crossfaded by situation (abyss, trainer, shadow_lugia, mewtwo, darkrai), seamless loop points, N to mute; nothing plays until the first click or key.
- Options (saved in the browser): every key binding, music volume, camera shake, simple graphics.
- URL flags: ?seed=text fixes the island, ?party=0025,0387 skips the title and roll, ?demo=1 the level-100 demo team (Infernape, shiny Rayquaza, Latios, shiny Jirachi, Tinkaton), ?moves=1 the Move Gallery, ?hoopa=1 forces Hoopa, ?dev=1 tester mode, ?rare=1 the rare-item chest, ?replay=match and ?profile=name open the game there. window.showdown exposes the live match in the console.

4.16 Characters (data/characters.json, Ranger sheets)

Player Ranger (union_00); rivals Rook (union_06), Marlo (union_08), Wynne (union_01), Brigg (union_02), Tamsin (union_04), Dale (union_07), Juniper (union_09); Ranger Union host (union_05); Castle Champion and Ranger Instructor (union_03); Nurse (civ_13); vendors (civ_01, civ_02, civ_03); outpost staff (civ_06, civ_09); Wandering Merchant (civ_07); Ace Trainer Celeste (civ_12); Breeder (civ_05); townsfolk drawn from civ_00, civ_03, civ_04,

civ_07, civ_08, civ_11, civ_12 and union_03/09; a Team Dim Sun grunt role exists with a placeholder (no sheet yet). Trainer looks: ten Ranger Union members and thirteen townsfolk sheets.

4.17 Every species in the game (322, in species.json order)

Turtwig, Chimchar, Piplup, Pikachu, Eevee, Riolu, Gible, Dratini, Larvitar, Starly, Bidoof, Shinx, Budew, Caterpie, Kricketot, Geodude, Krabby, Shellos, Magikarp, Psyduck, Wingull, Darkrai, Infernape, Rayquaza, Latios, Jirachi, Tinkaton, Grotle, Torterra, Monferno, Prinplup, Empoleon, Raichu, Vaporeon, Jolteon, Flareon, Espeon, Umbreon, Leafeon, Glaceon, Sylveon, Lucario, Gabite, Garchomp, Dragonair, Dragonite, Pupitar, Tyranitar, Staravia, Staraptor, Bibarel, Luxio, Luxray, Roselia, Roserade, Metapod, Butterfree, Kricketune, Graveler, Golem, Kingler, Gastrodon, Gyarados, Golduck, Pelipper, Snover, Abomasnow, Swinub, Piloswine, Mamoswine, Snorunt, Glalie, Spheal, Sealeo, Walrein, Sandshrew, Sandslash, Trapinch, Vibrava, Flygon, Cacnea, Cacturne, Hippopotas, Hippowdon, Hoothoot, Noctowl, Teddiursa, Ursaring, Vulpix, Ninetales, Sentret, Furret, Gastly, Haunter, Gengar, Bronzor, Bronzong, Honedge, Doublade, Aegislash, Grubbin, Charjabug, Vikavolt, Applin, Flapple, Appletun, Wurmple, Silcoon, Beautifly, Cascoon, Dustox, Tropius, Treecko, Grovyle, Sceptile, Ralts, Kirlia, Gardevoir, Gallade, Solosis, Duosion, Reuniclus, Fomantis, Lurantis, Impidimp, Morgrem, Grimmsnarl, Charcadet, Armarouge, Ceruledge, Deerling, Sawsbuck, Musharna, Zorua, Zoroark, Alolan Vulpix, Alolan Ninetales, Amaura, Aurorus, Absol, Frigibax, Arctibax, Baxcalibur, Avalugg, Sandile, Krokorok, Krookodile, Darumaka, Darmanitan, Steelix, Meowth, Persian, Elekid, Electabuzz, Electivire, Magby, Magmar, Magmortar, Smoochum, Jynx, Tyrogue, Hitmonlee, Hitmonchan, Hitmontop, Munchlax, Snorlax, Bulbasaur, Ivysaur, Venusaur, Charmander, Charmeleon, Charizard, Squirtle, Wartortle, Blastoise, Chikorita, Bayleef, Meganium, Cyndaquil, Quilava, Typhlosion, Totodile, Croconaw, Feraligatr, Torchic, Combusken, Blaziken, Mudkip, Marshtomp, Swampert, Snivy, Servine, Serperior, Tepig, Pignite, Emboar, Oshawott, Dewott, Samurott, Chespin, Quilladin, Chesnaught, Fennekin, Braixen, Delphox, Froakie, Frogadier, Greninja, Rowlet, Dartrix, Decidueye, Litten, Torracat, Incineroar, Popplio, Brionne, Primarina, Grookey, Thwackey, Rillaboom, Scorbunny, Raboot, Cinderace, Sobble, Drizzile, Inteleon, Sprigatito, Floragato, Meowscarada, Fuecoco, Crocalor, Skeledirge, Quaxly, Quaxwell, Quaquaval, Bagon, Shelgon, Salamence, Beldum, Metang, Metagross, Deino, Zweilous, Hydreigon, Goomy, Sliggoo, Goodra, Jangmo-o, Hakamo-o, Kommo-o, Dreepy, Drakloak, Dragapult, Mew, Celebi, Jirachi, Manaphy, Shaymin, Victini, Meltan, Marshadow, Cresselia, Lugia, Shadow Lugia, Ho-Oh, Hoopa, Suicune, Lucario, Garchomp, Dragonite, Tyranitar, Gyarados, Mega Charizard X, Mega Gengar, Mega Steelix, Mega Tyranitar, Mega Gardevoir, Mega Absol, Mega Glalie, Mega Latios, Mega Lucario, Mega Gallade, Tentacool, Tentacruel, Horsea, Seadra, Kingdra, Goldeen, Seaking, Lapras, Chinchou, Lanturn, Remoraid, Octillery, Wailmer, Wailord, Barboach, Whiscash, Feebas, Milotic, Finneon, Lumineon, Qwilfish, Corsola, Luvdisc, Relicanth, Raikou, Koraidon, Miraidon, Melmetal, Zacian, Zamazenta, Snom, Frosmoth, Cubchoo, Beartic, Bergmite, Sneasel, Weavile, Hoopa Unbound, Shadow Mewtwo, Darkrai

5. Online play (branch online)

- **Modes**: Casual (anyone, alone or in a party of up to four as a squad: teammates never fight, the last squad standing wins, knocked-out members included); Ranked (signed-in, solo; Glicko-2 extended to free-for-all placements, five placement games, tiers Bronze to Master, 60-day seasons from SEASON_START, everyone pulled halfway back to 1500 at a season start; bots count as weaker opponents and stop helping past 1750; lobbies match by rating with a widening window); Private lobbies (host makes one, shares the five-letter code; rules: trainers 2-8, bots on/off, starter level, teams solo/pairs/fours/pick, storm speed, the rare-item chest; nothing kept); Friends (requests by name, activity shown); Watching (a friend's match, the featured match, the rest of your own; 4 s behind; left/right switch the camera); Replays (every match recorded, what any trainer could see, keyframed every 30 s; Space pause, left/right 10 s, up/down speed, Q/E trainer; oldest dropped past REPLAY_MB 400).
- **Accounts**: trainer name and password (scrypt), optional Sign in with Google (only the account number and first name are used), eight one-time recovery codes instead of an email, sessions as HttpOnly Secure SameSite cookies, rate-limited sign-in, POST requests need a custom header. Guests can play but keep nothing. The privacy page is at /privacy; PRIVACY_CONTACT sets where deletion requests go.
- **Profiles**: in the game (click a name) and as a page to share (/u/name: level, title, rank, stats, achievements, recent matches with replay links).
- **Resilience**: a solo player's match survives reloads and wifi blips (45 s reconnect window), two tabs no longer kick each other (the newest wins), dead connections are detected both ways, the match screen says "Connection lost... reconnecting" and recovers or returns to the hub with a reason; nothing hangs forever.

- **Anti-cheat**: the server runs every match, so a modified browser cannot move faster, hit harder or spawn items; impossible commands are refused and flagged, enough flags in one match file a report. **Chat strikes**: a filtered word, flooding or repeating is a strike; three in ten minutes mutes for 10 minutes, then an hour, then a day. Known limit: everyone's position goes to every client for the minimap radar, so a traffic reader can scout.
- **Moderation**: the dashboard at /mod (open reports, a page per player with mute, kick, ban, devices and other accounts, history; the chat log; flagged players; live matches to watch; announcements; the log of every moderator action; the Server tab with players, matches, step time, bandwidth, memory, backups, grouped errors; Events: Double XP, shiny boost, faster or slower storm, timed announcements). The first administrator is made with node dist-server/admin.js admin Name over fly ssh.

6. Operations

- Fly.io app almia-showdown (live: shared-cpu-2x, 1 GB, LOBBY_WAIT 60, MATCH_SIZE 8, backups every 15 minutes, DRAIN_SECONDS 240, rare chest off) and almia-showdown-demo (fly.demo.toml: sleeps when idle, 20 s lobby, rare chest on, separate database). One machine only: the hub lives in memory and SQLite has one writer, so scale up, never out. Each match costs about 9% of a core and 2 KB/s per player; the live machine runs about six matches; shared-cpu-4x with 2 GB about fifteen; launch-day.md recommends performance-1x with 2 GB for a real launch.
- Deploys: the live deploy first stops new matches, waits up to 20 minutes for running ones, then deploys; whatever still runs when Fly stops the machine gets 4 more minutes, then is called off with nothing counted. OPS_TOKEN lets the workflow drain politely. The demo just deploys on every push to main.
- Backups: every 15 minutes the database is copied without pausing matches, gzipped, kept on the volume (last 8 and one a day for a week) and, with Tigris S3 set up, uploaded (live/latest, hourly, daily); one more on every shutdown; Fly snapshots the volume daily; restore with admin.js restore. Alerts (ALERT_WEBHOOK_URL) on each deploy, a new kind of error, a match crashing, the server falling behind for half a minute, backups failing twice. Metrics to Fly Grafana. GoatCounter counts page views and match-started, match-finished and match-won events. Health at /health, logs with fly logs.
- Settings (environment): LOBBY_WAIT, MATCH_SIZE, MIN_PLAYERS, DEV_CLUSTER, RARE_CHEST, RANKED_WAIT, SEASON_START, SEASON_DAYS, REPLAYS, REPLAY_MB, SIM_LATENCY_MS, ROOM_WORKERS, DATA_DIR, PUBLIC_URL, GOOGLE_CLIENT_ID and SECRET, OPS_TOKEN, ALERT_WEBHOOK_URL, PRIVACY_CONTACT.
- Locally: npm run server (:8787) and npx vite --port 5173, Play Online; npx tsx server/bots.ts 3 60; npx tsx server/headless.ts 120 3; bash scripts/check-server.sh runs every check against a built server.
- Still undone on the ops side: a real domain with certificates (PUBLIC_URL), sizing the machine for launch, confirming off-site backups and the Google secrets on the live app, a Discord alert webhook, phone controls, and the smoke test's known cosmetic failure ("first tab closed with code 4000" timing).

7. The island generator

The island is produced by a Python pipeline from The Fan-tasy Tileset (the pack lives outside the repo; set FANTASY_DIR to the folder holding Art/ and Tiled/). The rebuild order is: build3.py, polish3.py --write, a ground render, pave_smooth.py, a ground render with the overlay, center_blend.py, a full render, minimap_from_render.py.

- **design3.py**: the land (a rounded rectangle with noise, four lands by angle, the plain), plateaus (the foothill and the five mountain tiers in MOUNTAIN, set back seven rows at the south faces and four at the north, faces squared where the trail climbs at STAIR_SITES; desert mesas; the autumn ridge; the forest hill), lakes, Tide Lake, the river, the stream and waterfall, roads (ROADS with winding centre lines; climbing roads straight through their stairs), the castle rectangle and GATES, TOWNS, SUMMIT, CRYSTAL, FINAL_ARENA, PLAIN, BLIGHT.
- **paint3.py and wang.py**: terrain painted with the pack's own terrain sets (Tiled's terrain brush reproduced: cells to a lattice of corners and edges, then tiles by pattern with the artist's probabilities): grounds, sand, water, snow with winter grass showing through, leaves, roads, five kinds of tall grass by land. **cliffs.py and automap.py**: rock slopes

with the pack's automap rules (markers for body, upper side and stairs; ice on the snow mountain, grey, grey-sand and brown-snow elsewhere), stairs cut wherever a road meets a south face plus extra flights.

- **build3.py**: runs everything in order: towns (towns3.py: brick plaza, houses, stalls with vendors, well, props, the enclosure with gaps at roads, trees and a watchtower inside), the castle area, the crystal arena, cliffs, bridges where roads reach both banks of the river, farms (farm3.py), torches at plazas, the keep and stamps, decor (arena dressing, the mountain, the ruins, roads' signs and lamps and verge plants, lakes), woods and scatter (content3.py), the blight (dead trees, grey rocks, bones on the grey ground), shores, life (decor3.life: grass clumps, flower patches, reeds, lily pads, logs, rune rocks, braziers), waterfalls, tall grass, collisions from the pack's collider shapes (small things are see-through), walkable and reachable grids, NPC spots, outposts, then export3.py writes almia3.json with a tile atlas and object sheets.
- **castlekit.py** composes the castle from the Atlas_Castle_Building pieces (section 4.3); **castle3.py** turns it into an "area" (layers, solid, paving shapes, objects, markers) and lays out the garden; **areas.py** writes and reads the same area as a Tiled map (scripts/fantasy/areas/castle.tmx, all 131 tilesets referenced, preview layers rewritten after each build) so the castle can be edited in Tiled with no code; the round trip is exact. statue.py carves a stone Pokémon from its sprite. animprops.py finds the multi-tile animated props in the Animation_* sheets and places them in the Animations layer.
- **polish3.py**: the landscaper pass (groves, flower drifts, beach life, reeds, open-field singles) keeping off the castle, the oval and the blight. **pave_smooth.py**: paving edges redrawn at pixel level (true ovals, rounded corners, a rim, worn turf) into an overlay drawn under the Animations layer. **center_blend.py**: a per-pixel colour multiplier for the gradients where the plain meets snow, desert and autumn and for the ash-grey centre (grass only; tall grass keeps some green). **minimap_from_render.py**: the minimap and preview from the final render.
- Tileset use after this session: about 70 of the pack's 130 tilesets. Unused on purpose: the build-your-own house atlases (towns use the artist's finished houses), the city-wall set (its automap rules are incomplete and name a renamed tileset), the sky backgrounds.

The island as it is now (full render 5120 x 3584 px, one pixel per game pixel):





The ice mountain: five tiers, the trail on stairs with torches, the summit tower, camps.



The castle composed from the kit, with its garden courtyard and the battlefield oval.



The desert ruins by the oasis, boats on the shore.



The farm outside Greenhollow: field, crops, fence, farmhouse.

8. Work done in the most recent session (27 October 2026)

- Global chat moved to the right of the hub; trading between accounts, Wonder Trade, clickable chat names opening a trainer card with a Trade button; an Eliminations stat.
- PC sizing and centring fixed for every sprite, including the hover animation.
- A paid Crystal Healer near the arena (P$600); the healing outposts restyled (a nurse recoloured from a townsfolk sheet, a rippling white flag with a red cross).
- Hoopa enrages after three hits; a spawn-rate study (241 forms, about 218 wilds a match, Hoopa 15%); Hippowdon replaced by Hippopotas; lists of candidate desert and snow additions given but not chosen.
- Gradient transitions between the plain and the three bordering lands; then the centre drained to ash grey with dead things, confined to the grey ground; tall grass keeps some green.
- The ice plateau given more exits, the desert outpost moved north, roads widened to never under three tiles, roads converging on the arena at the four compass points, the snow roads winding again.
- The castle rebuilt on feedback: a stretch of the pack's Bosk Barracks sample, then a garden courtyard, then composed from the castle kit itself with kerbed lawns; the battlefield oval and the fountain with a stone Celebi kept throughout; the castle exported as an editable Tiled map referencing every tileset.
- Chests cut to 10 slots and the dev chest hidden behind ?rare=1; trainer levels switched off; bots only once half a lobby is human; the "empty seats get AI trainers" line removed; the online menu got a map backdrop; camera shake limited to the three legends; the Capture Styler made a rare held item with a cursor icon; a stuck-after-reload bug fixed (undefined fields now travel as deletions); the browser icon set to Wesley's pixel Lugia head.
- The island rebuilt with far more of the pack: the ice mountain (five tiers of ice cliffs with stairs, torches, the summit tower, braziers, banners, camps, a watchtower, flags, winter grass, woods, a waterfall), farms, a river with a stone bridge, town enclosures with trees, watchtowers and props, the desert ruins, boats and palms on the shores, torches at plazas, animated life everywhere, denser woods. The pack's documentation and sample maps were studied first; the sample maps and the full art were sent to Wesley.
- Every change was deployed to the live server and ported to the offline demo; all tests pass on both branches.

9. Conventions to keep

- Never use em dashes in any writing (code comments, commit messages, documents, chat).
- Commit messages end with: "Co-Authored-By: Claude Fable 5.1 <noreply@anthropic.com>" and "Claude-Session: https://claude.ai/code/session_019L7iVyevzrSsZHNd1CijFN".
- Work on branch online, cherry-pick to main (drop server/ and online-only files on conflict), merge main back into online with the ours strategy, deploy with the Launch online server workflow, confirm the run finished.
- Keep Hoopa, Shadow Lugia and Shadow Mewtwo out of marketing material.
- A Fly.io org token and 2FA recovery codes were once pasted in chat; they were to be deleted and regenerated and must never be reused or stored.
- Build from the pack's kits and individual assets (terrain sets, automap rules, object collections, animation sheets), not by cutting pieces out of its sample maps. Every tile and object must exist in the pack.
- The README's "The map" section still describes the old Sunnyside-based map and should be rewritten for the Fan-tasy island.

10. Open items and likely next steps

- Choose the desert and snow spawn additions (ready now: Snorunt, Swinub, Bergmite, Smoochum, Lapras; Sandshrew, Cacnea, Geodude, Gible; others need sprites fetched with scripts/fetch-sprites.py).
- More for the centre arena from the pack's assets (the user asked for a cooler arena).

- Towns built from the build-your-own house atlases (a house grammar from the Atlas_Buildings kits); the remaining pack sets (city walls, sky backgrounds, 1-tile rock slopes); towns and the arena as editable Tiled areas like the castle.
- Public-launch operations: domain and certificates, machine size, off-site backups, Google secrets, alert webhook, phone controls, the two-tab 4000 message.
- Decide whether the final showdown should be on (duel.finalShowdown).

11. Key numbers

Thing	Value
Island	320 x 224 tiles of 16 px (5120 x 3584 px); 25 ground layers; 2,172 placed objects; 18 NPC spots; 5 healing outposts; 4 towns; 2 farms; 1 river with a bridge; 1 waterfall
Content	322 species (11 Mega forms, shiny and shadow forms as palette swaps and extra folders), 175 moves, 64 items, 38 TMs, 18 types, 16 townsfolk, 25 achievements, 5 music tracks, 23 trainer looks
Code	about 42,000 lines (TypeScript and Python), 294 commits; 81 unit tests on online, 56 on main
Match	8 trainers, 60 steps a second, about 218 wilds, darkness every 200 s in five pulses, 1.5 ms of CPU per step on the server
Server	Fly.io shared-cpu-2x 1 GB live, a demo app that sleeps; about 9% of a core and 2 KB/s per player per match
Pack	The Fan-tasy Tileset: 130 tilesets, 4,088 art files, 16 sample maps; about 70 tilesets in use